

# Abhishek Kumar Singh

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## Professional Summary

Motivated B.Tech Computer Science student at BIT Mesra (**CGPA 8.91**) with hands-on AI/ML and data analytics experience . Built production-grade Power BI dashboards integrating Python ML models and NLP on real enterprise data. Developed and deployed an empathetic LLM counseling chatbot using LoRA/PEFT fine-tuning, FastAPI, and Docker on Hugging Face Spaces. Skilled across the full ML lifecycle — data wrangling, model training, evaluation, and cloud deployment. **GATE CS & DS 2026** qualifier.

## Education

**B.Tech — Computer Science And Engineering – CGPA: 8.91**

2023 - 2027

*Birla Institute of Technology, Mesra*

**Higher Secondary (10+2) — 94.6%**

2020 - 2022

*Holy Mary International School · Darbhanga, Bihar*

## Experience

### Data Analytics Intern

May 2026 – July 2026

*Oil and Natural Gas Corporation Limited (ONGC) · Mumbai, Maharashtra*

- ▶ Engineered a **5-page interactive Power BI dashboard** (Executive, SLA Analysis, Department Analytics, Sentiment Intelligence, AI Insights) processing **~2,000 real IT helpdesk tickets** for ONGC leadership.
- ▶ Integrated Python ML models — **Random Forest classifier** for priority prediction and **Gradient Boosting regressor** for resolution-time forecasting — directly into the Power BI pipeline via Python visuals.
- ▶ Performed end-to-end data wrangling in **Power Query (M language)**: null imputation, casing normalization, date parsing, SLA breach flag derivation; reduced data inconsistencies by **~35%**.
- ▶ Authored **25+ DAX measures** including SLA compliance rate, MTTR, ticket velocity, and agent performance index enabling real-time executive drill-through analysis.
- ▶ Applied **NLP-based sentiment scoring** (VADER / TextBlob) on ticket descriptions, surfacing department-level frustration trends used in monthly operations reviews.
- ▶ Delivered ONGC-branded reports reviewed and evaluated by senior mentors as the internship capstone deliverable.

## Projects

### EmpathAI — Empathetic LLM Counseling Chatbot

2026

*Python · Hugging Face · LangChain · FastAPI · Docker · Streamlit · MLflow*

- ▶ Fine-tuned **Mistral-7B using LoRA/PEFT** on curated mental-health conversation datasets, improving domain-specific empathy scores by **38%** over the base model in human evaluation.
- ▶ Designed a **multi-turn conversation pipeline with LangChain**, maintaining session context and implementing crisis-detection guardrails with safe fallback responses.
- ▶ Built a **REST API backend with FastAPI**, containerized with **Docker**, and deployed on **Hugging Face Spaces + Render**; achieved sub-2s inference latency on free-tier GPU.
- ▶ Validated with a 50-person pilot: **82% positive response quality** rating; reduced hallucination rate by **40%** vs. baseline through system-prompt engineering and **RAG (Retrieval-Augmented Generation)**.
- ▶ Implemented **MLflow experiment tracking** to log hyperparameters, loss curves, and ROUGE/BERTScore metrics across fine-tuning runs.

### AI Code Review Bot

2026

*Python · FastAPI · LLM · RAG · ChromaDB · AST · Static Analysis · GitHub*

- ▶ Built an **AI-powered code review engine** combining LLM semantic analysis, Git diff parsing, AST analysis, and static-analysis evidence to identify security vulnerabilities, bugs, performance issues, and code-quality problems.
- ▶ Developed a modular **FastAPI backend** with structured **Pydantic/JSON output validation, confidence-based filtering, prompt engineering, and LLM provider abstraction** for reliable, evidence-grounded reviews.
- ▶ Implemented a **ChromaDB-based RAG knowledge base** containing curated security, Python, API, testing, and coding guidelines to provide relevant context for AI-generated recommendations.
- ▶ Created comprehensive **Pytest automation** covering AI review logic, malformed LLM responses, confidence filtering, AST analysis, Git diff parsing, API validation, static analysis, and LLM failure handling.

## Technical Skills

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**Languages:** Python, Java, C, C++, SQL, JavaScript, HTML/CSS

**ML / AI:** Scikit-learn, Random Forest, Gradient Boosting, XGBoost, TensorFlow, PyTorch (basics), Transformers (Hugging Face), LoRA/PEFT fine-tuning, RAG

**Data & BI:** Power BI (DAX, Power Query M), Pandas, NumPy, Matplotlib, Jupyter Notebook

**MLOps & Deployment:** FastAPI, Docker, MLflow, Hugging Face Spaces, Render, Git, GitHub Actions (CI/CD basics), REST APIs

**Databases:** MySQL, MongoDB

**Tools & IDEs:** VS Code, Anaconda, Google Colab

## Achievements

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- ▶ **GATE CS & DS 2026** — Qualified national-level exam, demonstrating strong proficiency in CS fundamentals, Data Structures, and Algorithms.
  - ▶ Shortlisted for ONGC Data Analytics Internship (2025) from a competitive pool of CSE students across multiple universities.
  - ▶ Published a literature survey on AI-driven mental health chatbots analyzing 20+ research papers on culturally sensitive LLM-based CBT approaches.
  - ▶ **Sports Excellence** — Secured Runner-up in both Badminton and Cricket at inter-branch college tournaments, demonstrating teamwork and consistency.
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